Air Brake Department

Inspection and Testing the E. T. Locomotive Brake in Passenger Service—Must Conform to the Rules of the Interstate Commerce Commission

In order to make the series of questions and answers that refer to locomotive brake inspection more comprehensive, it may be well to outline in a general way, the manner in which the brakes may be properly inspected and tested in the shortest possible time. Assuming a passenger locomotive of ordinary construction, it is well to proceed in the following manner.

Start the inspection with approximately 45 lbs pressure in the brake cylinders, the independent brake valve in lap position and the automatic valve in running position. Then start brake cylinder leakage test by closing the stop cock in the distributing valve supply pipe.

Beginning under the engine at the pilot, examine all brake cylinders and pipe connections with a torch, for leakage, measure piston travel, see that all cylinders and cylinder bolts are securely tightened and that no part of foundation brake rigging is less than 2½ ins. above the rail. Report brake shoes that are not approximately in line with the tread of the wheel, note that all piping is properly clamped and free from leakage, drain any main reservoir under engine and note last date of hydrostatic test, see that it is securely fastened, test air hose between engine and tender following air pipes to the rear of tender, giving the tender brake cylinder the same attention as accorded driving brake cylinders.

From the rear of the tender, go to distributing valve side of engine, testing this valve for leakage from exhaust port or gasket, test main, equalizing and distributing valve reservoirs for leakage and see that all are securely tightened, drain right main reservoir, noting last date of hydrostatic test, and follow brake and signal pipes to pilot. Test air hose and piping between hose and stop cocks on pilot, pass to left main reservoir testing it and related piping for leakage, and open drain cock.

Examine air compressor, as will be specified in the questions and answers, also the water pipe from the compressor governor. When this operation has been completed, close reservoir drain cock, blow dirt out of hose couplings at rear of tender and attach a test gage and device with various sized openings next to the signal hose, and test signal whistle with a 1/16 in. and 3/64 in. opening, and if necessary to cut larger sized opening as will be explained. The feed valve is then to be tested with both a 3/64 in. and a 1/16 in. circular opening and with one larger sized opening for a capacity test.

After feed valve test has been completed, close orifices and carefully note figure of feed valve adjustment and immediately enter the cab to compare test gage with engine gages. This may be varied if two inspectors are used on the same engine or if special apparatus is used whereby a test gage may be brought into the cab window of the locomotive.

Place automatic brake valve handle on lap position and close the air pump throttle, test all piping in cab, signal valve, brake valve gaskets and exhaust ports for leakage, see that all valves are tight on brackets, note last date of brake cleaning, air compressor and air gage tests and test air pump governor and operating pipe for leakage.

Note the drop in air pressure, as shown by the gages during the interval the brake valve remained on lap position and the pump throttle was closed. Leakage in the main reservoir and pipe connections must not exceed 2 lbs. per minute, if it does another test of 3 minutes duration must be made after the pressure has reduced 60 per cent below the maximum carried. Leakage from the brake pipe volume must not exceed 3 lbs. per minute.

At this time 5 minutes or more will have elapsed since the inspection was started, no pressure remaining in the brake cylinders, brake must remain applied for a period of not less than 5 minutes with communication to the brake cylinders closed.

Open air pump throttle, move brake valve to running position, and open the stop cock in the distributing valve supply pipe.

When air pressures have been restored to the maximum, move automatic brake valve to service position, making a 5 lb. brake pipe reduction, which should apply the brakes, unless the distributing valve is of a special type. During this time note that the equalizing discharge valve of the brake valve responds to a light reduction in equalizing reservoir pressure and closes off tightly when the reduction ceases and that the compressor governor is sensitive enough to permit the compressor to start promptly when the brake applies.

From lap position again move the brake valve to service position, making 5 lbs. more brake pipe reduction, which should result in 25 lbs. brake cylinder pressure for the total 10 lbs. reduction. Make 10 lbs. more to see that 50 lbs. cylinder pressure is obtained for 20 lbs. reduction in brake pipe pressure. Make 20 lbs. more brake pipe reduction and note that brake cylinder pressure does not exceed 68 lbs., and follow with another reduction that will bring brake pipe pressure to a lower figure than brake cylinder pressure to see that there is no back leakage from the brake cylinders into the brake pipe, which would then unseat the equalizing discharge valve of the brake valve.

Release brake with the independent brake valve, application cylinder pressure should be practically exhausted in from 2 to 3 seconds' time.

Move automatic brake valve to release position, until pressure chamber of the distributing valve is charged to 125 or 130 lbs. During this time main reservoir, equalizing reservoir and brake pipe pressure will register the same if air gages are correct, as determined by the test gage when entering the cab, after the feed valve test.

When pressure chamber is sufficiently overcharged make another reduction with the automatic brake valve just sufficient to apply the brake, and with the brake pipe pressure still above 110 lbs., return brake valve to running position. The brake should remain applied.

With the automatic brake valve reduce brake pipe pressure to 110 lbs., then direct from 110 to 90 lbs. and the equalizing reservoir reduction must take place in from 5½ to 6 seconds time. Any variation in this rate of reduction must be reported.

Return automatic brake valve handle to holding position, the feed valve must promptly return the brake pipe pressure to 110 lbs., and the brake must remain applied (standard equipment). Test the release pipe branch between the brake valves for leakage.

Move the automatic brake valve to running position, and after brake has completely released, move independent brake valve to slow application position, 40 lbs. brake cylinder pressure should be obtained in from 6 to 8 seconds' time.

Graduate the brake off by alternating the valve handle between lap and running position, and if the distributing valve is sufficiently sensitive brake cyl-

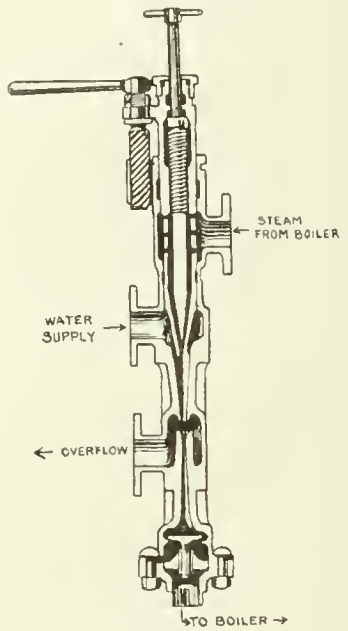
Construction and Operation of the Injector

Giffard's Invention—Subsequent Improvements—Injector Troubles and Their Remedies

When the injector was originally introduced in 1858 by Giffard, a French engineer, the idea of water being forced into a boiler against the same pressure as the motive steam was received with ridicule. This did not hinder its adoption or its subsequent improvement. The explanation in the light of experience, shows that the velocity with which a fluid or gas such as air or steam issues from a vessel is greater for a given pressure, the lighter the fluid or gas. In discharging water or steam from a boiler, the velocity of steam compared to water under the same pressure is, on the average, twenty-five times as great. At 60 lbs. pressure steam has a velocity of 2,400 feet per second, water 94 feet. This ratio is nearly maintained almost equally in all the varying pressures of modern boiler service. It is not surprising therefore that the fluid with the excessive velocity will penetrate the comparatively slow moving even at the same pressure per square inch. Of course the mixture of water and steam in the injector greatly reduces the velocity, for example, one pound of steam at 120 lbs. pressure, having a velocity of 2,850 feet per second, is mixed with ten pounds of water, the resulting velocity of the combined jet of water and condensed steam would be 2,850 ÷ 11 = 259 feet per second. Water alone at the same pressure would have a velocity of 133 feet per second, so that the jet of steam and water has an ample excess of velocity to overcome the opposing jet of water alone from the boiler. An additional velocity of forty to fifty feet per second is quite sufficient to overcome the added resistance of valves and piping necessary for the appliance of the injector.

This is not all. The form of the nozzles are important. At first the steam nozzle was simply that of a long, gradually tapering convergent tube. Much of the energy was lost by the sudden expansion of the jet of steam. At present nearly all modern injectors have steam nozzles tapering towards the discharge end. In Giffard's original invention a regulating spindle was introduced, arranged in such a way that it could be moved up and down thus varying the entrance to the combining tube. In its original form it gave constant trouble. Improvements followed, but the regulation of the openings both for steam and water by hand required a certain amount of skill, and with a view to render the adjustments automatic, in 1864 Mr. Sellers, of Philadelphia, brought out the self-regulating injector, which under modified and improved forms has reached a stable degree of perfection.

In our own day it may be said that the injectors are all as nearly perfect as human ingenuity and continued experiment can approach perfection. Injector failures are rare occurrences, and it may be set down as a fact that injector troubles

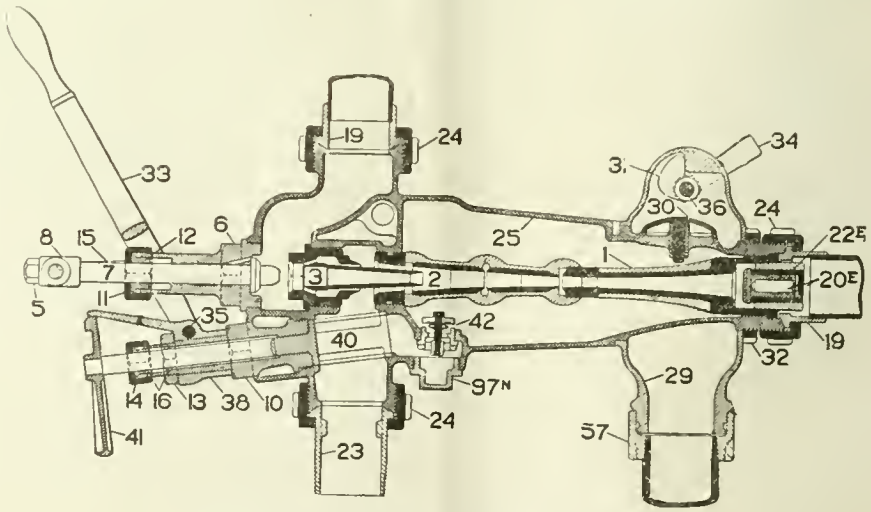


GIFFARD'S ORIGINAL TYPE OF INJECTOR.

are not to be attributed to organic defects.

There is much more likelihood in troubles arising with the injector connections than there is about the injector itself. The pipes leading to and from the injector should never be smaller in size than the injector connection. Nearly all injectors used on locomotives are known as the lifting type. In order that the injector may be conveniently reached in the cab of the locomotive it is usually placed higher than the water in the tank, and consequently the water must be lifted the amount of difference between the surface of the water and the injector opening. In order to lift the water, a vacuum is produced in the pipe leading from the tank to the injector, by admitting steam into the injector and allowing it to pass through into the overflow pipe. The interior of the injector is so constructed that the steam finds its way into the overflow pipe much more readily than it does into the pipe leading to the tank. The air in the pipe leading to the tank is induced to mix with the steam and pass out at the overflow pipe, while the weight of the atmosphere on the surface of the water in the tank forces the water into the vacuum thus produced. The joints forming the connection between the injector and the tank should be of the most secure kind, as a leak in this pipe will greatly diminish, if it does not altogether stop the operation of the injector. In some experiments where injectors are placed below the level of the water in the tank the improvement in the working of the injector is of a marked kind.

In all injector troubles the failure of the water to reach the injector is the most common, and if there is water in the tank and the surface be not frozen, as it may readily be in winter, the trouble will likely be by reason of a leak in the suction pipe or by clogging of the strainers. A heated suction pipe also often refuses to supply water in sufficient quantity for the effective use of the injector. In cases where the injector lifts the water and fails to force it into the boiler, the trouble should be readily evident by the behavior of the injector. If the steam should blow back into the tank it will either be caused by reason of the overflow valve being partly closed, thereby



SELLER'S IMPROVED SELF-ACTING INJECTOR.

Train Handling.

(*Continued from page* 264, *August*, 1917.)

35. Q.—How would the brake valve be handled in coupling to a string of uncharged cars if a hurried movement was to be made before the brake test?

A.—By making one or two heavy applications with the automatic brake valve and leaving the valve handle on lap position while the hose connections are made.

36. Q.—For what purpose?

A.—To have a reduced pressure in the pressure chamber of the distributing valve reservoir or in any auxiliary reservoir on the engine or tender, so that there will be no brake sticking.

37. Q.—Is there any other advantage in this?

A.—This will prevent a full application of the engine brakes when the hose are coupled and thus avoid an additional waste of main reservoir pressure through the brake cylinders, and the maximum main reservoir pressure will be accumulated for the charging of the train.

38. Q.—What position of the brake valve will then be used to charge the train?

A.—Full release position.

39. Q.—When should the brake valve handle thereafter be brought to running position?

A.—When the brake pipe pressure is about up to the adjustment of the feed valve.

40. Q.—Why do the compressors stop if the brake valve handle is brought to running position before the auxiliary reservoirs are charged?

A.—Because the auxiliary reservoirs absorb brake pipe pressure at a faster rate than at which the main reservoir pressure can flow through the feed valve into the brake pipe.

41. Q.—How does this cause the compressors to stop?

A.—The difference between main reservoir and brake pipe pressure will exceed the tension of the regulating spring of the excess pressure governor top, and the diaphragm valve of this governor top will become unseated and stop the compressors.

42. Q.—How long will the compressors remain shut down?

A.—Until the main reservoir pressure reduces, through expansion into the brake pipe, to a figure within 20 lbs. of the pressure in the brake pipe.

43. Q.—How is it ascertained that the train is about charged?

A.—By the slowing down of the compressors and the positions of the air gage hands.

44. Q.—How is a test for brake pipe leakage made?

A.—By noting the fall of the brake pipe gage hand after an application of the brakes has been made.

45. Q.—If the brake valve is placed in lap position and the gage hand of the brake pipe falls, how does the action of the brakes indicate whether the fall in pressure is due to brake pipe leakage or an uncharged train?

A.—If it is due to brake pipe leakage the brakes will apply and brake pipe pressure will continue to fall, but if some of the auxiliary reservoirs are still uncharged, the brakes will only apply on the cars with the charged reservoirs and brake pipe pressure will cease to fall when equalization has taken place.

46. Q.—After waiting a reasonable length of time for the reservoirs to become charged, what could be wrong if the compressors stopped every time the brake valve handle was brought to running position?

A.—That brake pipe leakage was in excess of the capacity of the feed valve.

47. Q.—What would be done if the train crew was to state that there was no unusual leakage in the brake pipe?

A.—A test of the capacity of the brake pipe feed valve would be made.

48. Q.—How may this be done?

A.—By closing the brake valve cut out cock and open the brake pipe of the engine to the atmosphere; then with maximum air pressure in the main reservoir and brake pipe above the cut out cock, open the cut out cock and note the fall of main reservoir pressure.

49. Q.—With the main reservoir capacity usually found on modern locomotives, that is, from 60,000 to 80,000 cubic inches, with the brake pipe feed valve in good condition and no restrictions in the brake pipe, how long will it require for the pressure in the main reservoir to reduce 20 lbs. under this condition?

A.—It should reduce from 90 to 70 lbs. in from 30 to 40 seconds, and in passenger service from 130 to 110 lbs. in from 15 to 20 seconds' time.

50. Q.—What would you think wrong if the governor would not hold the compressors shut down during this test; that is, with the brake pipe open to the atmosphere?

A.—That there was some restriction in the brake pipe or brake valve of the locomotive, and that because of it, feed valve pipe pressure was being maintained.

51. Q.—Is there anything else that could cause the compressors to stop besides brake pipe leakage in excess of the capacity of the feed valve?

A.—Yes, a defect of the excess pressure top of the pump governor.

52. Q.—What would you do if the feed valve was working at its maximum capacity during this test; that is, expanding the main reservoir pressure into the brake pipe in the specified length of time?

A.—Examine the piping of the locomotive for leakage, and insist that the leaks in the brake pipe be tightened to a degree that the feed valve will be able to maintain the pressure in the brake pipe.

53. Q.—What would be done if it required approximately one minute for the main reservoir pressure to reduce from 9 to 70 lbs., with the size of reservoirs specified?

A.—The engine would be taken back to the shop unless specific instructions forbid this.

54. Q.—Why not clean the feed valve when indications are that it is defective?

A.—The trouble is not likely to be in the feed valve, and even if it usually requires a drill press to remove obstructions from the feed valve and sometimes necessitates the removal of the supply valve piston bushing.

55. Q.—Why not proceed with the brake valve in release position?

A.—The train might be parted en route, and with large capacity compressors the pressure on the engine may be held to a standard with the angle cock at the rear end of the train open to the atmosphere, so that with the engine brake released with the independent brake valve, the carrying of the brake valve in release position might result in a portion of the train being left along the road.

56. Q.—Is there any other reason why a train should not be run over the road with brake pipe leakage that is in excess of the capacity of the feed valve?

A.—Yes, with excessive brake pipe leakage the control of the train is taken away from the engineer.

57. Q.—In what manner?

A.—If instructions call for a specified amount of brake pipe reduction for a stop, and brake pipe leakage continues the reduction, instructions are not being followed and the brakes are not under the control of the engineer.

58. Q.—How should an attempt to release brakes be made in the event of brakes sticking?

A.—By a quick movement of the brake valve to release and back to running position.

59. Q.—Why a quick movement?

A.—To "kick off" any brakes that may have applied through an inequality of pressure, without overcharging the auxiliary reservoirs at the head end of the train.

60. Q.—Should a few of these movements to release be made just before ascending a grade to insure that brakes are released?

A.—Under no circumstances, except in actual cases of brakes sticking.

61. Q.—What would several of these movements to release likely accomplish?

A.—An overcharge of the auxiliary reservoirs at the heads end.

(*To be continued*)